

Q1 2021 (01 Sep Shift 2)

The width of one of the two slits in a Young's double slit experiment is three times the other slit. If the amplitude of the light coming from a slit is proportional to the slit-width, the ratio of minimum to maximum intensity in the interference pattern is $x : 4$ where x is

Q2 2021 (31 Aug Shift 2)

In a Young's double slit experiment, the slits are separated by 0.3 mm and the screen is 1.5 m away from the plane of slits. Distance between fourth bright fringes on both sides of central bright is 2.4 cm. The frequency of light used is $\times 10^{14}$ Hz

Q3 2021 (27 Aug Shift 2)

The light waves from two coherent sources have same intensity $I_1 = I_2 = I_0$. In interference pattern the intensity of light at minima is zero. What will be the intensity of light at maxima?

- (1) I_0
- (2) $2I_0$
- (3) $5I_0$
- (4) $4I_0$

Q4 2021 (26 Aug Shift 2)

A source of light is placed in front of a screen. Intensity of light on the screen is I . Two Polaroids P_1 and P_2 are so placed in between the source of light and screen that the intensity of light on screen is $I/2$. P_2 should be rotated by an angle of (degrees) so that the intensity of light on the screen becomes $\frac{3I}{8}$

Q5 2021 (26 Aug Shift 1)

White light is passed through a double slit and interference is observed on a screen 1.5 m away. The separation between the slits is 0.3 mm. The first violet and red fringes are formed 2.0 mm and 3.5 mm away from the central white fringes. The difference in wavelengths of red and violet light isnm

Answer Key

Q1 (1)

Q2 (5)

Q3 (4)

Q4 (3)

Q5 (300)

Q1 (1)

Given amplitude $\propto$ slit widthAlso intensity $\propto (\text{Amplitude})^2 \propto (\text{Slit width})^2$

$$\frac{I_1}{I_2} = \left(\frac{3}{1}\right)^2 = 9 \Rightarrow I_1 = 9I_2$$

$$\frac{I_{\min}}{I_{\max}} = \left(\frac{\sqrt{I_1} - \sqrt{I_2}}{\sqrt{I_1} + \sqrt{I_2}}\right)^2 = \left(\frac{3-1}{3+1}\right)^2 = \frac{1}{4} = \frac{x}{4}$$

$$\Rightarrow x = 1.00$$

Q2 (5)

$$8\beta = 2.4 \text{ cm}$$

$$\frac{8\lambda\Delta}{d} = 2.4 \text{ cm}$$

$$\frac{8 \times 1.5 \times c}{0.3 \times 10^{-3} \times f} = 2.4 \times 10^{-2}$$

$$f = 5 \times 10^{14} \text{ Hz}$$

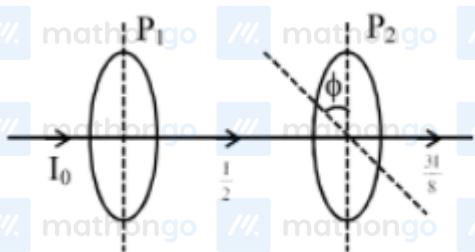
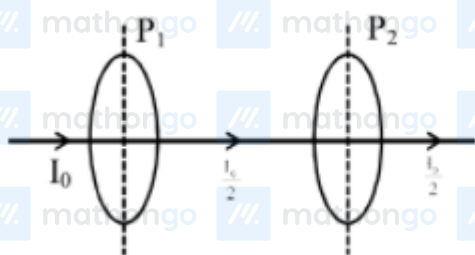
Q3 (4)

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$= 4I_0$$

Q4 (3)

$$I = \frac{I_0}{2} \cos^2 \phi$$



$$\frac{I}{2} \cos^2 \phi = \frac{3I}{8}$$

$$\cos^2 \phi = \frac{3}{4}$$

$$\cos^2 \phi = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \phi = 30$$

Ans. 30

Q5 (300)

Position of bright fringe $y = n \frac{D\lambda}{d}$

$$y_1 \text{ of red} = \frac{D\lambda_r}{d} = 3.5 \text{ mm}$$

$$\lambda_r = 3.5 \times 10^{-3} \frac{d}{D}$$

Similarly $\lambda_v = 2 \times 10^{-3} \frac{d}{D}$

$$\lambda_r - \lambda_v = (1.5 \times 10^{-3}) \left(\frac{0.3 \times 10^{-3}}{1.5} \right)$$

$$= 3 \times 10^{-7} = 300 \text{ nm}$$

Ans. 300.0